

Roll No. ....

**B.Sc. (Hons.) SEMESTER IV EXAMINATION 2023-24****MATHEMATICS****MTB - 401 : Partial Differential Equations****Time : Three hours****Max. Marks : 70****(WRITE YOUR ROLL NO. AT THE TOP IMMEDIATELY ON THE RECEIPT OF THIS QUESTION PAPER)***The figures in the right hand margin indicate marks.***Answer five questions including Question 1 which is compulsory.****1. Attempt all parts of this question**

- (a) Form the partial differential equation by eliminating
- $h$
- and
- $k$
- from the equation:

$$(x - h)^2 + (y - k)^2 + z^2 = \lambda^2$$

[2]

- (b) Form a partial differential equation by eliminating arbitrary functions
- $f$
- and
- $g$
- from the equation:

$$y = f(x + at) + g(x - at).$$

[2]

- (c) Solve the p.d.e.
- $z \frac{\partial z}{\partial x} = -y$
- .
- [2]

- (d) Find the complete integral of
- $z = px + qy + p^2 + q^2$
- , where
- $p = \partial z / \partial x$
- ,
- $q = \partial z / \partial y$
- .
- [2]

- (e) Solve
- $(D_x^3 - 4D_x^2 D_y + 4D_x D_y^2)z = 0$
- , where
- $D_x \equiv \partial / \partial x$
- ,
- $D_y \equiv \partial / \partial y$
- .
- [3]

- (f) Find the general solution of
- $s + p - q = z$
- .
- [3]

- (g) Solve
- $x^2 r - y^2 t + xp - yq = 0$
- by reducing it in linear partial differential equation with constant coefficients.
- [3]

- (h) Solve
- $xr + p = 9x^2 y^2$
- .
- [3]

- (i) Check whether the partial differential equation
- $3r + 6s + 4t = 0$
- is elliptic.
- [2]

**2. (a) Obtain the general integral of the partial differential equation**

$$p + 3q = 5z + \tan(y - 3x).$$

[5]

- (b) Apply Charpit's method to find the complete integral of the equation
- $px + qy + z = xq^2$
- .
- [7]

**3. (a) Find the general integral of the equation  $r + s - 6t = y \cos x$ . [5]**

- (b) Reduce the equation

$$x^2 \frac{\partial^2 z}{\partial x^2} - 4xy \frac{\partial^2 z}{\partial x \partial y} + 4y^2 \frac{\partial^2 z}{\partial y^2} + 6y \frac{\partial z}{\partial y} = x^3 y^4$$

to linear equation with constant coefficients and then solve it. [7]

(2)

4. (a) Solve  $(2D_x^2 - 5D_xD_y + 2D_y^2)z = 24(y - x)$ . [5]  
(b) Solve the non-homogeneous partial differential equation

$$(D_x^2 - D_xD_y - 2D_y^2 + 2D_x + 2D_y)z = e^{3x+2y}.$$

[7]

5. (a) Solve:  $yx - p = xy^2 \cos(xy)$ . [5]  
(b) Reduce the following equation to canonical form and solve it

$$y^2 \frac{\partial^2 u}{\partial x^2} - 2xy \frac{\partial^2 u}{\partial x \partial y} + x^2 \frac{\partial^2 u}{\partial y^2} = \frac{y^2}{x} \frac{\partial u}{\partial x} + \frac{x^2}{y} \frac{\partial u}{\partial y}.$$

[7]

6. (a) Solve the following partial differential equation by Monge's method:

$$y^2 r - 2yz + t = p + 6y.$$

[5]

- (b) Solve:  $r + 4s + t + rt - s^2 = 2$  [7]

7. (a) Use the method of separation of variables to solve the heat equation

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \quad 0 < x < 1, t > 0,$$

subject to the boundary conditions  $u(x, 0) = x^2 - x$ ,  $u(0, t) = u(1, t) = 0$ . [6]

- (b) Solve the wave equation

$$\frac{\partial^2 u}{\partial t^2} = a^2 \frac{\partial^2 u}{\partial x^2}$$

using the method of separation of variables. [6]

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#### External Portals & Academic Resources:

**Examination Portal:** <https://bhu-exam.vercel.app/>

**Student Resource Center:** <https://quashan-me.vercel.app/>

**Academic Portfolio:** <https://me-aryan.vercel.app>